MATH1211

Chapter 6: Integration

ABU UBAEDA

Type 1: Trigonometric Substitution & Special Trigonometric Forms

Problem 1

$$\int \frac{d\theta}{5 + 4\cos\theta}$$

Solution: Use the standard half-angle substitution: Let $t = \tan\left(\frac{\theta}{2}\right) \Longrightarrow d\theta = \frac{2\,dt}{1+t^2}$ and $\cos\theta = \frac{1-t^2}{1+t^2}$.

Substitute these into the integral:

$$\int \frac{\frac{2\,dt}{1+t^2}}{5 + 4\left(\frac{1-t^2}{1+t^2}\right)} = \int \frac{2\,dt}{5(1+t^2) + 4(1-t^2)} = \int \frac{2\,dt}{9+t^2}$$

Using the standard integral formula $\int \frac{du}{a^2+u^2} = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + C$:

$$\int \frac{2\,dt}{3^2+t^2} = \frac{2}{3}\arctan\left(\frac{t}{3}\right) + C = \frac{2}{3}\arctan\left(\frac{\tan(\theta/2)}{3}\right) + C$$

Problem 2

$$\int \frac{dx}{\cos x + \sin x}$$

Solution: Express the denominator as a single trigonometric term using the identity $a\cos x + b\sin x = R\cos(x-\phi)$:

$$\cos x + \sin x = \sqrt{2}\left(\frac{1}{\sqrt{2}}\cos x + \frac{1}{\sqrt{2}}\sin x\right) = \sqrt{2}\cos\left(x - \frac{\pi}{4}\right)$$

Substitute back into the integral:

$$\int \frac{dx}{\sqrt{2}\cos\left(x-\frac{\pi}{4}\right)} = \frac{1}{\sqrt{2}}\int \sec\left(x-\frac{\pi}{4}\right)dx$$

Using $\int \sec(u)\,du = \ln|\sec u + \tan u| + C$:

$$= \frac{1}{\sqrt{2}}\ln\left|\sec\left(x-\frac{\pi}{4}\right) + \tan\left(x-\frac{\pi}{4}\right)\right| + C$$

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Problem 3

$$\int \frac{dx}{1+\cos\alpha\cos x}$$

Solution: Use the identities $1=\sin^2\left(\frac{x}{2}\right)+\cos^2\left(\frac{x}{2}\right)$ and $\cos x=\cos^2\left(\frac{x}{2}\right)-\sin^2\left(\frac{x}{2}\right)$:

$$1+\cos\alpha\cos x=\cos^2\left(\frac{x}{2}\right)(1+\cos\alpha)+\sin^2\left(\frac{x}{2}\right)(1-\cos\alpha)$$

Divide numerator and denominator by $\cos^2\left(\frac{x}{2}\right)$:

$$\int \frac{\sec^2\left(\frac{x}{2}\right)dx}{(1+\cos\alpha)+(1-\cos\alpha)\tan^2\left(\frac{x}{2}\right)}$$

Let $u=\tan\left(\frac{x}{2}\right)\Longrightarrow du=\frac{1}{2}\sec^2\left(\frac{x}{2}\right)dx\Longrightarrow 2\,du=\sec^2\left(\frac{x}{2}\right)dx$:

$$=\int \frac{2\,du}{(1+\cos\alpha)+(1-\cos\alpha)u^2}=\frac{2}{1-\cos\alpha}\int \frac{du}{\left(\frac{1+\cos\alpha}{1-\cos\alpha}\right)+u^2}$$

Using $1+\cos\alpha=2\cos^2\left(\frac{\alpha}{2}\right)$ and $1-\cos\alpha=2\sin^2\left(\frac{\alpha}{2}\right)$:

$$\frac{1+\cos\alpha}{1-\cos\alpha}=\cot^2\left(\frac{\alpha}{2}\right)$$

$$=\csc^2\left(\frac{\alpha}{2}\right)\int \frac{du}{\cot^2\left(\frac{\alpha}{2}\right)+u^2}=\csc^2\left(\frac{\alpha}{2}\right)\cdot\frac{1}{\cot(\alpha/2)}\arctan\left(\frac{u}{\cot(\alpha/2)}\right)+C$$

$$=\frac{2}{\sin\alpha}\arctan\left(\tan\left(\frac{\alpha}{2}\right)\tan\left(\frac{x}{2}\right)\right)+C$$

Problem 4

$$\int \frac{2\sin x+3\cos x}{3\sin x+4\cos x}dx$$

Solution: Express Numerator $=l$(Denominator) $+m$(Derivative of Denominator):

$$2\sin x+3\cos x=l(3\sin x+4\cos x)+m(3\cos x-4\sin x)$$

$$2\sin x+3\cos x=(3l-4m)\sin x+(4l+3m)\cos x$$

Equating coefficients:

1. $3l-4m=2$

2. $4l + 3m = 3$

Solving this system yields $l = \frac{18}{25}$ and $m = \frac{1}{25}$.

Rewrite the integral:

$$\int \frac{\frac{18}{25}(3\sin x + 4\cos x) + \frac{1}{25}(3\cos x - 4\sin x)}{3\sin x + 4\cos x} dx$$

$$= \frac{18}{25}\int dx + \frac{1}{25}\int \frac{3\cos x - 4\sin x}{3\sin x + 4\cos x} dx$$

$$= \frac{18}{25}x + \frac{1}{25}\ln|3\sin x + 4\cos x| + C$$

Problem 5

$$\int \frac{2 + 3\sin x - \cos x}{1 + \cos x + \sin x} dx$$

Solution: Express Numerator $= l$(Derivative of Denominator) $+ m$(Denominator) $+ n$:

$$2 + 3\sin x - \cos x = l(-\sin x + \cos x) + m(1 + \cos x + \sin x) + n$$

$$2 + 3\sin x - \cos x = (m - l)\sin x + (l + m)\cos x + (m + n)$$

Equating coefficients:

- $m - l = 3$
- $l + m = -1$
- $m + n = 2$

Solving gives $l = -2$, $m = 1$, and $n = 1$.

Substitute back:

$$\int \left[\frac{-2(-\sin x + \cos x)}{1 + \cos x + \sin x} + 1 + \frac{1}{1 + \cos x + \sin x}\right] dx$$

$$= -2\ln|1 + \cos x + \sin x| + x + \int \frac{dx}{1 + \cos x + \sin x}$$

For the last integral, convert to half angles:

$$\int \frac{dx}{2\cos^2(x/2) + 2\sin(x/2)\cos(x/2)} = \frac{1}{2}\int \frac{\sec^2(x/2)}{1 + \tan(x/2)} dx = \ln\left|1 + \tan\left(\frac{x}{2}\right)\right|$$

Final result:

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$$= x - 2\ln|1 + \cos x + \sin x| + \ln\left|1 + \tan\left(\frac{x}{2}\right)\right| + C$$

Problem 6

$$\int_0^{\pi/2} \frac{a\cos x + b\sin x}{a\sin x + b\cos x}\,dx$$

Solution: Using the definite integral property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$: Let $I = \int_0^{\pi/2} \frac{a\cos x + b\sin x}{a\sin x + b\cos x}\,dx$.

Applying the property:

$$I = \int_0^{\pi/2} \frac{a\cos(\frac{\pi}{2} - x) + b\sin(\frac{\pi}{2} - x)}{a\sin(\frac{\pi}{2} - x) + b\cos(\frac{\pi}{2} - x)}\,dx = \int_0^{\pi/2} \frac{a\sin x + b\cos x}{a\cos x + b\sin x}\,dx$$

Adding the two expressions for I:

$$2I = \int_0^{\pi/2} \left[\frac{a\cos x + b\sin x}{a\sin x + b\cos x} + \frac{a\sin x + b\cos x}{a\cos x + b\sin x}\right] dx$$

$$2I = \int_0^{\pi/2} \frac{(a\cos x + b\sin x)^2 + (a\sin x + b\cos x)^2}{(a\sin x + b\cos x)(a\cos x + b\sin x)}\,dx$$

Alternatively, splitting numerator as l(Denom) + m(Deriv):

$$a\cos x + b\sin x = l(a\sin x + b\cos x) + m(a\cos x - b\sin x)$$

Yields $l = \frac{2ab}{a^2+b^2}$ and $m = \frac{a^2-b^2}{a^2+b^2}$.

Evaluating directly:

$$I = \frac{\pi}{4} + \frac{a^2 - b^2}{2(a^2 + b^2)} \ln\left(\frac{a}{b}\right)$$

Type 2: Algebraic Substitutions & Radical Forms

Problem 7

$$\int \frac{dx}{\sqrt{1 + x^2}}$$

Solution: Let $x = \tan\theta \implies dx = \sec^2\theta\, d\theta$ and $\sqrt{1 + x^2} = \sec\theta$:

$$\int \frac{\sec^2\theta\, d\theta}{\sec\theta} = \int \sec\theta\, d\theta = \ln|\sec\theta + \tan\theta| + C$$

Substitute back $\tan\theta = x$ and $\sec\theta = \sqrt{1+x^2}$:

$$= \ln\left|x + \sqrt{1+x^2}\right| + C$$

Problem 8

$$\int \sqrt{\frac{a+x}{a-x}}\, dx$$

Solution: Rationalize the numerator:

$$\int \sqrt{\frac{(a+x)(a+x)}{(a-x)(a+x)}}\, dx = \int \frac{a+x}{\sqrt{a^2-x^2}}\, dx = a\int \frac{dx}{\sqrt{a^2-x^2}} + \int \frac{x\, dx}{\sqrt{a^2-x^2}}$$

1. $a\int \frac{dx}{\sqrt{a^2-x^2}} = a\arcsin\left(\frac{x}{a}\right)$
2. For $\int \frac{x\, dx}{\sqrt{a^2-x^2}}$, let $t = a^2 - x^2 \Longrightarrow dt = -2x\, dx$:

$$\int \frac{-\frac{1}{2}dt}{\sqrt{t}} = -\sqrt{t} = -\sqrt{a^2-x^2}$$

Combine both parts:

$$= a\arcsin\left(\frac{x}{a}\right) - \sqrt{a^2-x^2} + C$$

Problem 9

$$\int \frac{dx}{x\sqrt{1-x^2}}$$

Solution: Let $x = \sin\theta \Longrightarrow dx = \cos\theta\, d\theta$:

$$\int \frac{\cos\theta\, d\theta}{\sin\theta\sqrt{1-\sin^2\theta}} = \int \frac{\cos\theta}{\sin\theta\cos\theta}\, d\theta = \int \csc\theta\, d\theta$$

$$= -\ln|\csc\theta + \cot\theta| + C$$

Substitute back $\csc\theta = \frac{1}{x}$ and $\cot\theta = \frac{\sqrt{1-x^2}}{x}$:

$$= -\ln\left|\frac{1+\sqrt{1-x^2}}{x}\right| + C$$

Type 3: Partial Fraction Decomposition

Problem 10

$$\int \frac{x^2+1}{(x+1)(x-1)}dx$$

Solution: Simplify the improper fraction:

$$\frac{x^2+1}{x^2-1} = 1 + \frac{2}{x^2-1}$$

Decompose $\frac{2}{(x-1)(x+1)}$ using partial fractions:

$$\frac{2}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1} \Longrightarrow 2 = A(x+1) + B(x-1)$$

- Setting $x = 1 \Longrightarrow A = 1$
- Setting $x = -1 \Longrightarrow B = -1$

Integrate:

$$\int \left(1 + \frac{1}{x-1} - \frac{1}{x+1}\right)dx = x + \ln|x-1| - \ln|x+1| + C = x + \ln\left|\frac{x-1}{x+1}\right| + C$$

Problem 11

$$\int \frac{x-1}{(x-2)(x-3)}dx$$

Solution: Decompose using partial fractions:

$$\frac{x-1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3} \Longrightarrow x - 1 = A(x-3) + B(x-2)$$

- Setting $x = 2 \Longrightarrow A = -1$
- Setting $x = 3 \Longrightarrow B = 2$

Integrate:

$$\int \left(\frac{2}{x-3} - \frac{1}{x-2}\right)dx = 2\ln|x-3| - \ln|x-2| + C$$

Problem 12

$$\int \frac{x^2}{(x+1)(x+2)^2}\,dx$$

Solution: Decompose into partial fractions:

$$\frac{x^2}{(x+1)(x+2)^2} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$$

$$x^2 = A(x+2)^2 + B(x+1)(x+2) + C(x+1)$$

- Setting $x = -1 \Longrightarrow A = 1$
- Setting $x = -2 \Longrightarrow C = -4$
- Equating x^2 coefficients: $A + B = 1 \Longrightarrow 1 + B = 1 \Longrightarrow B = 0$

Integrate:

$$\int \left(\frac{1}{x+1} - \frac{4}{(x+2)^2}\right) dx = \ln|x+1| + \frac{4}{x+2} + C$$

Problem 13

$$\int \frac{x^2}{(x^2+1)(x^2+4)}\,dx$$

Solution: Let $u = x^2$ for partial fraction decomposition:

$$\frac{u}{(u+1)(u+4)} = \frac{A}{u+1} + \frac{B}{u+4} \Longrightarrow u = A(u+4) + B(u+1)$$

- Setting $u = -1 \Longrightarrow A = -\frac{1}{3}$
- Setting $u = -4 \Longrightarrow B = \frac{4}{3}$

Substitute back $u = x^2$:

$$\int \left(-\frac{1}{3}\frac{1}{x^2+1} + \frac{4}{3}\frac{1}{x^2+4}\right) dx$$

$$= -\frac{1}{3}\arctan(x) + \frac{4}{3}\cdot\frac{1}{2}\arctan\left(\frac{x}{2}\right) + C = \frac{2}{3}\arctan\left(\frac{x}{2}\right) - \frac{1}{3}\arctan(x) + C$$

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Problem 14

$$\int \frac{x}{(x^2+1)(x+1)}dx$$

Solution: Decompose into partial fractions:

$$\frac{x}{(x^2+1)(x+1)} = \frac{Ax+B}{x^2+1} + \frac{C}{x+1}$$

$$x = (Ax+B)(x+1) + C(x^2+1)$$

- Setting $x = -1 \Longrightarrow -1 = 2C \Longrightarrow C = -\frac{1}{2}$
- Equating x^2 coefficients: $A + C = 0 \Longrightarrow A = \frac{1}{2}$
- Equating constant terms: $B + C = 0 \Longrightarrow B = \frac{1}{2}$

Rewrite and integrate:

$$\int \left(\frac{\frac{1}{2}x + \frac{1}{2}}{x^2+1} - \frac{\frac{1}{2}}{x+1} \right) dx$$

$$= \frac{1}{4}\int \frac{2x}{x^2+1}dx + \frac{1}{2}\int \frac{dx}{x^2+1} - \frac{1}{2}\int \frac{dx}{x+1}$$

$$= \frac{1}{4}\ln(x^2+1) + \frac{1}{2}\arctan(x) - \frac{1}{2}\ln|x+1| + C$$

Category 1: Fundamental Theorems

Problem 1: Fundamental Theorem of Calculus

Statement 1: If f is continuous on a closed interval $[a, b]$, and $g(x)$ is defined by $g(x) = \int_a^x f\,(t)\,dt$, then $g(x)$ is continuous on $[a, b]$, differentiable on (a, b), and:

$$g'(x) = \frac{d}{dx}\left[\int_a^x f\,(t)\,dt\right] = f(x)$$

Statement 2: If f is continuous on $[a, b]$ and F is any antiderivative of f, then:

$$\int_a^b f\,(x)\,dx = F(b) - F(a)$$

Category 2: Area Under and Between Curves

Problem 2: Area bounded by $y^2 = ax$ and $x^2 + y^2 = 2ax$

Find the area bounded between the parabola $y^2 = ax$ and the circle $x^2 + y^2 = 2ax$.

Step 1: Points of intersection Substitute $y^2 = ax$ into the circle's equation:

$$x^2 + ax = 2ax \Longrightarrow x^2 - ax = 0 \Longrightarrow x(x - a) = 0$$

Points of intersection are $(0,0)$ and (a, a).

Step 2: Set up the integral Upper curve: $y_1 = \sqrt{2ax - x^2}$ Lower curve: $y_2 = \sqrt{ax}$

$$A = \int_0^a (y_1 - y_2)\, dx = \int_0^a \sqrt{2ax - x^2}\ dx - \int_0^a \sqrt{ax}\ dx$$

Step 3: Evaluate each part

1. For $\int_0^a \sqrt{2ax - x^2}\ dx$:

   Rewrite as $\int_0^a \sqrt{a^2 - (x - a)^2}\ dx$. Let $x - a = a\sin\theta \Longrightarrow dx = a\cos\theta\, d\theta$. Limits: $x = 0 \Longrightarrow \theta = -\frac{\pi}{2}$, $x = a \Longrightarrow \theta = 0$.

   $$\int_{-\pi/2}^{0} a\cos\theta \cdot a\cos\theta\, d\theta = a^2 \int_{-\pi/2}^{0} \frac{1 + \cos 2\theta}{2}\, d\theta = \frac{\pi a^2}{4}$$

2. For $\int_0^a \sqrt{ax}\ dx$:

   $$\sqrt{a}\left[\frac{2}{3}x^{3/2}\right]_0^a = \frac{2}{3}a^2$$

Final Result:

$$A = \frac{\pi a^2}{4} - \frac{2}{3}a^2 = a^2\left(\frac{\pi}{4} - \frac{2}{3}\right)$$

Problem 3: Area of an Ellipse

Find the total area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution: By symmetry across all four quadrants:

$$A = 4\int_0^a y\ dx = 4\frac{b}{a}\int_0^a \sqrt{a^2 - x^2}\ dx$$

Let $x = a\sin\theta \Longrightarrow dx = a\cos\theta\, d\theta$. Limits: $x = 0 \rightarrow \theta = 0$, $x = a \rightarrow \theta = \frac{\pi}{2}$.

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$$A = \frac{4b}{a}\int_0^{\pi/2} (a\cos\theta)(a\cos\theta)\, d\theta = 4ab\int_0^{\pi/2} \cos^2\theta\, d\theta$$

$$A = 2ab\int_0^{\pi/2} (1+\cos 2\theta)\, d\theta = 2ab\left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\pi/2} = \pi ab$$

Problem 4: Area bounded by $y^2 = 4x$ and $y = 2x$

Find the area bounded between $y^2 = 4x$ and $y = 2x$.

Solution:

1. **Points of intersection:**

   $(2x)^2 = 4x \Longrightarrow 4x^2 - 4x = 0 \Longrightarrow 4x(x-1) = 0 \Longrightarrow x = 0,1$. Intersection points: $(0,0)$ and $(1,2)$.

2. **Setup and evaluate:**

$$A = \int_0^1 (2\sqrt{x} - 2x)\, dx = 2\left[\frac{x^{3/2}}{3/2}\right]_0^1 - 2\left[\frac{x^2}{2}\right]_0^1 = \frac{4}{3}(1) - 1(1) = \frac{1}{3}$$

Problem 5: Area bounded by $y^2 = x^3$ and $y = x$

Find the area between $y^2 = x^3$ (i.e., $y = x^{3/2}$) and $y = x$.

Solution:

1. **Points of intersection:**

   $x^3 = x^2 \Longrightarrow x^2(x-1) = 0 \Longrightarrow x = 0,1$. Points: $(0,0)$ and $(1,1)$.

2. **Setup and evaluate:**

$$A = \int_0^1 (x - x^{3/2})\, dx = \left[\frac{x^2}{2} - \frac{2}{5}x^{5/2}\right]_0^1 = \frac{1}{2} - \frac{2}{5} = \frac{1}{10}$$

Problem 6: Area bounded by $x^2 + y^2 = 1$ and $y^2 = 1 - x$

Find the bounded area between the circle $x^2 + y^2 = 1$ and the parabola $y^2 = 1 - x$.

Solution:

1. **Intersection Points:**

$x^2 + (1 - x) = 1 \Longrightarrow x^2 - x = 0 \Longrightarrow x = 0,1$. Points: $(0, \pm 1)$ and $(1,0)$.

2. **Integration along the y-axis:**

$$A = 2\int_0^1 \left[(1 - y^2) - \sqrt{1 - y^2}\right] dy$$

$$= 2\left[y - \frac{y^3}{3}\right]_0^1 - 2\int_0^1 \sqrt{1 - y^2}\ dy = 2\left(\frac{2}{3}\right) - 2\left(\frac{\pi}{4}\right) = \frac{4}{3} - \frac{\pi}{2}$$

Problem 7: Area of an Astroid

Find the total area enclosed by $x^{2/3} + y^{2/3} = a^{2/3}$.

Solution:

1. **Symmetry:**

   $A = 4\int_0^a y\ dx = 4\int_0^a \left(a^{2/3} - x^{2/3}\right)^{3/2} dx.$

2. **Parametric Substitution:**

   Let $x = a\sin^3\theta \Longrightarrow dx = 3a\sin^2\theta\cos\theta\ d\theta$. Limits: $x = 0 \to \theta = 0, x = a \to \theta = \frac{\pi}{2}$.

3. **Evaluate:**

$$A = 4\int_0^{\pi/2} (a\cos^3\theta) \cdot (3a\sin^2\theta\cos\theta)\, d\theta = 12a^2 \int_0^{\pi/2} \sin^2\theta\cos^4\theta\, d\theta$$

   Using Wallis' Formula:

$$\int_0^{\pi/2} \sin^2\theta\cos^4\theta\, d\theta = \frac{(1)(3 \cdot 1)}{(2 + 4)(4)(2)} \cdot \frac{\pi}{2} = \frac{3}{32} \cdot \frac{\pi}{2} = \frac{3\pi}{64}$$

$$A = 12a^2\left(\frac{\pi}{32}\right) = \frac{3}{8}\pi a^2$$

Problem 8: Area of a Hypocycloid

Find the area enclosed by $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$.

Solution: Using substitution $x = a\sin^3\theta, y = b\cos^3\theta$:

$$A = 12ab\int_0^{\pi/2}\sin^2\theta\cos^4\theta\,d\theta = 12ab\left(\frac{\pi}{32}\right) = \frac{3}{8}\pi ab$$

Problem 9: Common Area between Ellipses $x^2 + 2y^2 = a^2$ and $2x^2 + y^2 = a^2$

Solution:

1. **Intersection:**

   Subtracting gives $y^2 - x^2 = 0 \Longrightarrow y = \pm x$. Substituting $y = x \Longrightarrow 3x^2 = a^2 \Longrightarrow x = \pm\frac{a}{\sqrt{3}}$.

2. **Total Area calculation by 8-fold symmetry:**

$$A = 8\int_0^{a/\sqrt{3}}\sqrt{a^2 - 2x^2}\,dx = 8\sqrt{2}\int_0^{a/\sqrt{3}}\sqrt{\left(\frac{a}{\sqrt{2}}\right)^2 - x^2}\,dx$$

$$= 8\sqrt{2}\left[\frac{x}{2}\sqrt{\frac{a^2}{2} - x^2} + \frac{a^2}{4}\arcsin\left(\frac{x\sqrt{2}}{a}\right)\right]_0^{a/\sqrt{3}}$$

$$= 8\sqrt{2}\left(\frac{a}{2\sqrt{3}}\frac{a}{\sqrt{6}} + \frac{a^2}{4}\arcsin\sqrt{\frac{2}{3}}\right) = \frac{4}{3}a^2 + 2\sqrt{2}a^2\arcsin\sqrt{\frac{2}{3}}$$

Problem 10: Common Area between $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

Solution: Intersection in the first quadrant occurs along line $y = x$. Substituting $y = x$ gives $x = \frac{ab}{\sqrt{a^2+b^2}}$. By 8-fold symmetry, the total common area evaluates to:

$$A = 2ab\arctan\left(\frac{2ab}{a^2 - b^2}\right)$$

Category 3: Arc Length & Perimeter

Problem 11: Arc Length of $y^2 = 4ax$

Find the arc length of the parabola $y^2 = 4ax$ from the vertex $(0,0)$ to an extremity of the latus rectum $(a, 2a)$.

Solution:

1. **Derivative:**

   $2y\frac{dy}{dx} = 4a \Longrightarrow \frac{dy}{dx} = \frac{2a}{y} = \sqrt{\frac{a}{x}}.$

2. **Arc Length Formula:**

$$S = \int_0^a \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \int_0^a \sqrt{1 + \frac{a}{x}}\, dx = \int_0^a \sqrt{\frac{x+a}{x}}\, dx$$

3. **Substitution:**

   Let $x = u^2 \Longrightarrow dx = 2u\, du$. Limits: $x = 0 \to u = 0, x = a \to u = \sqrt{a}$.

$$S = \int_0^{\sqrt{a}} \sqrt{\frac{u^2 + a}{u^2}}\,(2u\, du) = 2\int_0^{\sqrt{a}} \sqrt{u^2 + a}\; du$$

$$= 2\left[\frac{u}{2}\sqrt{u^2 + a} + \frac{a}{2}\ln\left(u + \sqrt{u^2 + a}\right)\right]_0^{\sqrt{a}}$$

$$= a\left[\sqrt{2} + \ln\left(1 + \sqrt{2}\right)\right]$$

Problem 12: ***Perimeter of a Circle*** $x^2 + y^2 = a^2$

Find the perimeter using parametric equations.

Solution:

1. **Parametric Equations:** $x = a\cos t, y = a\sin t$ for $t \in [0, 2\pi]$.
2. **Derivatives:** $\frac{dx}{dt} = -a\sin t, \frac{dy}{dt} = a\cos t.$
3. **Perimeter Formula:**

$$P = \int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = \int_0^{2\pi} \sqrt{a^2\sin^2 t + a^2\cos^2 t}\; dt$$

$$P = \int_0^{2\pi} a\; dt = a[t]_0^{2\pi} = 2\pi a$$

Problem 13: Arc Length of Catenary $y = \frac{a}{2}\left(e^{x/a} + e^{-x/a}\right)$

Find the arc length from the vertex $(0, a)$ to any point (x_1, y_1).

Solution:

1. **Derivative:**

$$\frac{dy}{dx} = \frac{1}{2}\left(e^{x/a} - e^{-x/a}\right) = \sinh\left(\frac{x}{a}\right)$$

2. **Arc Length Integral:**

$$S = \int_0^{x_1} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \int_0^{x_1} \sqrt{1 + \sinh^2\left(\frac{x}{a}\right)}\, dx = \int_0^{x_1} \cosh\left(\frac{x}{a}\right) dx$$

$$S = \left[a\sinh\left(\frac{x}{a}\right)\right]_0^{x_1} = a\sinh\left(\frac{x_1}{a}\right) = \frac{a}{2}\left(e^{x_1/a} - e^{-x_1/a}\right)$$

Problem 14: Perimeter of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$

1. Parametric equations:

$$x = a\cos^3\theta, \quad y = a\sin^3\theta, \quad 0 \le \theta \le 2\pi$$

2. Derivatives:

$$\frac{dx}{d\theta} = -3a\cos^2\theta\sin\theta, \quad \frac{dy}{d\theta} = 3a\sin^2\theta\cos\theta$$

3. Differential arc length ds:

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = 9a^2\cos^4\theta\sin^2\theta + 9a^2\sin^4\theta\cos^2\theta = 9a^2\sin^2\theta\cos^2\theta(\cos^2\theta + \sin^2\theta)$$

$$\Rightarrow \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = 3a\sin\theta\cos\theta$$

4. Total Perimeter: By four-fold symmetry across the four quadrants ($0 \le \theta \le \frac{\pi}{2}$):

$$P = 4\int_0^{\pi/2} 3\,a\sin\theta\cos\theta\, d\theta = 12a\left[\frac{\sin^2\theta}{2}\right]_0^{\pi/2} = 12a\left(\frac{1}{2}\right) = 6a$$

Problem 15: Perimeter of the hypo-cycloid $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$

1. Parametric equations:

$$x = a\cos^3\theta, \quad y = b\sin^3\theta, \quad 0 \le \theta \le 2\pi$$

2. Derivatives:

$$\frac{dx}{d\theta} = -3a\cos^2\theta\sin\theta, \quad \frac{dy}{d\theta} = 3b\sin^2\theta\cos\theta$$

3. Differential arc length ds:

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = 9\sin^2\theta\cos^2\theta(a^2\cos^2\theta + b^2\sin^2\theta)$$

$$ds = 3\sin\theta\cos\theta\sqrt{a^2\cos^2\theta + b^2\sin^2\theta}\, d\theta$$

4. Total Perimeter:

$$P = 4\int_0^{\pi/2} 3\sin\theta\cos\theta\sqrt{a^2\cos^2\theta + b^2\sin^2\theta}\, d\theta$$

Let $u = a^2\cos^2\theta + b^2\sin^2\theta$. Then $du = 2(b^2 - a^2)\sin\theta\cos\theta\, d\theta \Longrightarrow \sin\theta\cos\theta\, d\theta = \frac{du}{2(b^2-a^2)}$.
When $\theta = 0 \Longrightarrow u = a^2$; when $\theta = \frac{\pi}{2} \Longrightarrow u = b^2$.

$$P = 12\int_{a^2}^{b^2} \sqrt{u}\frac{du}{2(b^2 - a^2)} = \frac{6}{b^2 - a^2}\left[\frac{2}{3}u^{3/2}\right]_{a^2}^{b^2} = \frac{4(b^3 - a^3)}{b^2 - a^2} = \frac{4(a^2 + ab + b^2)}{a + b}$$

Problem 16: Perimeter of the evolute $(ax)^{2/3} + (by)^{2/3} = (a^2 - b^2)^{2/3}$

Let $C = (a^2 - b^2)^{2/3}$. Rewrite as:

$$\left(\frac{ax}{C^{3/2}}\right)^{2/3} + \left(\frac{by}{C^{3/2}}\right)^{2/3} = 1$$

This is a scaled hypo-cycloid with effective semi-axes $a' = \frac{a^2-b^2}{a}$ and $b' = \frac{a^2-b^2}{b}$. Using the perimeter formula derived in 1(iv):

$$P = \frac{4(a'^2 + a'b' + b'^2)}{a' + b'}$$

Substituting a' and b':

$$P = \frac{4(a^2 - b^2)\left(\frac{1}{a^2} + \frac{1}{ab} + \frac{1}{b^2}\right)}{\frac{1}{a} + \frac{1}{b}} = \frac{4(a^2 - b^2)\left(\frac{a^2 + ab + b^2}{a^2b^2}\right)}{\frac{a+b}{ab}} = \frac{4(a-b)(a^2 + ab + b^2)}{ab}$$
$$= \frac{4(a^3 - b^3)}{ab}$$

Problem 17: Arc length of $ay^2 = x^3$ from the cusp $(0,0)$ to (x, y)

1. Differentiate:

$$y = \frac{x^{3/2}}{a^{1/2}} \Rightarrow \frac{dy}{dx} = \frac{3}{2}\sqrt{\frac{x}{a}}$$

2. Arc Length Integral:

$$s = \int_0^x \sqrt{1 + \left(\frac{dy}{dt}\right)^2}\, dt = \int_0^x \sqrt{1 + \frac{9t}{4a}}\, dt$$

Let $u = 1 + \frac{9t}{4a} \Rightarrow du = \frac{9}{4a}dt$:

$$s = \frac{4a}{9}\left[\frac{2}{3}\left(1 + \frac{9t}{4a}\right)^{3/2}\right]_0^x = \frac{8a}{27}\left[\left(1 + \frac{9x}{4a}\right)^{3/2} - 1\right]$$

Problem 18: Show $3s^2 = 4x^2 + 3y^2$ for $3ay^2 = x(x-a)^2$

1. Differentiate the curve equation:

$$3ay^2 = x(x-a)^2 = x^3 - 2ax^2 + a^2x$$

$$6ay\frac{dy}{dx} = 3x^2 - 4ax + a^2 = (3x - a)(x - a)$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{(3x-a)^2(x-a)^2}{36a^2y^2}$$

Since $3ay^2 = x(x-a)^2 \Rightarrow 36a^2y^2 = 12a \cdot 3ay^2 = 12ax(x-a)^2$:

$$\left(\frac{dy}{dx}\right)^2 = \frac{(3x-a)^2(x-a)^2}{12ax(x-a)^2} = \frac{(3x-a)^2}{12ax}$$

2. Set up $1 + \left(\frac{dy}{dx}\right)^2$:

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{9x^2 - 6ax + a^2}{12ax} = \frac{9x^2 + 6ax + a^2}{12ax} = \frac{(3x+a)^2}{12ax}$$

$$\Rightarrow \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{3x + a}{\sqrt{12ax}}$$

3. Compute Arc Length s:

$$s = \int_0^x \frac{3t + a}{\sqrt{12at}}\, dt = \frac{1}{2\sqrt{3a}} \int_0^x \left(3t^{1/2} + at^{-1/2}\right) dt = \frac{1}{2\sqrt{3a}} \left[2t^{3/2} + 2at^{1/2}\right]_0^x$$

$$s = \frac{\sqrt{x}(x + a)}{\sqrt{3a}}$$

4. Verify identity $3s^2 = 4x^2 + 3y^2$:

- Left side:

$$3s^2 = 3\left(\frac{x(x + a)^2}{3a}\right) = \frac{x(x + a)^2}{a}$$

- Right side:

$$4x^2 + 3y^2 = 4x^2 + \frac{x(x - a)^2}{a} = \frac{4ax^2 + x(x^2 - 2ax + a^2)}{a} = \frac{x^3 + 2ax^2 + a^2x}{a} = \frac{x(x + a)^2}{a}$$

Since both sides match, $3s^2 = 4x^2 + 3y^2$ is proved.

Problem 19: Compare perimeters of $\frac{x^2}{9} + \frac{y^2}{7} = 1$ and $\frac{x^2}{36} + \frac{y^2}{28} = 1$

1. Perimeter formula for an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$: The perimeter P is given by $P = 4aE(e)$, where $e = \sqrt{1 - \frac{b^2}{a^2}}$ is the eccentricity, and $E(e)$ is the complete elliptic integral of the second kind.

2. Analyze Ellipse 1 ($\frac{x^2}{9} + \frac{y^2}{7} = 1$):

- $a_1 = 3, \quad b_1 = \sqrt{7}$
- $e_1 = \sqrt{1 - \frac{7}{9}} = \sqrt{\frac{2}{9}} = \frac{\sqrt{2}}{3}$
- $P_1 = 4(3)E\left(\frac{\sqrt{2}}{3}\right) = 12E\left(\frac{\sqrt{2}}{3}\right)$

3. Analyze Ellipse 2 ($\frac{x^2}{36} + \frac{y^2}{28} = 1$):

- $a_2 = 6, \quad b_2 = \sqrt{28} = 2\sqrt{7}$
- $e_2 = \sqrt{1 - \frac{28}{36}} = \sqrt{\frac{8}{36}} = \frac{\sqrt{2}}{3}$
- $P_2 = 4(6)E\left(\frac{\sqrt{2}}{3}\right) = 24E\left(\frac{\sqrt{2}}{3}\right)$

Conclusion:

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$$P_2 = 2P_1$$

The perimeter of the second conic is exactly **twice** the perimeter of the first conic.

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